LABORATORY-SCALE PLANETARY SURFACE MAPPING USING USRP-BASED SYNTHETIC APERTURE RADAR

This mini-project demonstrates a laboratory-scale implementation of planetary surface mapping using microwave radar principles. A USRP software-defined radio and horn antenna are employed to transmit Linear Frequency Modulated pulse (LFM) signals toward various target surfaces and receive the corresponding reflected echoes. The received signals are processed to estimate target range, relative reflectivity, and surface roughness characteristics. By linearly translating the antenna across a predefined path and coherently combining multiple radar measurements, a two-dimensional radar reflectivity image is formed using Synthetic Aperture Radar (SAR) techniques. The project highlights fundamental radar concepts and spaceborne radar imaging used in planetary exploration missions. This experimental setup offers a cost-effective platform for understanding radar-based surface characterization and SAR imaging principles.

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Design and Implemetation of Compact 77 GHz Synthetic Aperture Radar for Drone Based Applications

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***Abstract*—The demands on more immediately operating synthetic aperture radar (SAR) systems are increasing as SAR became a universal remote sensor that is applicable to a variety of missions. SAR systems mounted on unmanned aerial vehicles (UAV) are expected to provide suitable solutions to such demands, as extensive technological advancements have been made over the past decade in related fields. This paper presents a compact SAR system integrated into a rotary wing consumer drone. The system is designed into the dual receiver single-input-multiple-output (SIMO) configuration in order to provide a rich applicability for the system, including the capability of single path interferometry. To overcome inherent restrictions of small UAV platforms, millimeter wave architecture is adopted to provide considerable compactness and efficiency for the system. Design details and results from an outdoor experiment which was performed to prove that the system can produce proper SAR data are discussed as well.**

Keywords—Synthetic Aperture Radar, SAR, Millimeter Wave, Drone, UAV

I. Introduction

The remote sensing via synthetic aperture radar (SAR) is meeting its golden age, as numerous studies have proven that SAR sensors are highly effective to a variety of important missions including land deformation detection, forest and agricultural survey, maritime surveillance, earth topography mapping and so on [1]-[3]. However, since the majority of active SAR sensors nowadays are spaceborne, and targeting the global coverage, the revisit time issue arises for certain missions (e.g. safety monitoring over man-made structures, detection of time-sensitive targets) which require immediate and frequent observations. For missions with high locality and frequent revisit, an airborne SAR system mounted on the manned aircraft were preferred. Nevertheless, considering maintenance expenses and requirement for a skilled pilot, only a huge organizations, such as the military or space agency have the ability to operate traditional airborne SAR systems. This hinders the wide-spread of SAR applications over civil uses.

Along with the extensive technological advancements made in the field of unmanned aerial vehicle (UAV) over the past decade, SAR systems based on UAV platforms were proposed to be an appropriate solution for such problem. UAVs, especially those with multi rotor configuration are capable of vertical take-off and automatized navigation via GNSS observations. Such characteristics offer a great advantage to missions that require instant deployment of sensors, or frequent and periodic survey on same region of interest. Several studies present a successful development of UAV based SAR systems, dedicated to a various civil purposes [4]-[7].

In this paper, we present a decent SAR system mounted on multirotor UAV platform that is based on the compact millimeter wave architecture. The paper is organized as follows: We'll provide detailed descriptions on system design parameters and specifications of each hardware of proposed system in Section II. Finally, in Section III, we'll provide an experimental results performed to verify the imaging capability of the system, along with the discussion on SAR synthesis procedure we've adopted.



Fig. 1. SAR Instrument Fully Integrated with Platform and Peripherals.

II. System Description

A. 77 GHz W – Band RadarRF Front End

In the presented system, linear frequency modulated continuous wave (FMCW) signal whose center frequency is 77 GHz and bandwidth of 1 GHz is transmitted. The major benefit of adopting millimeter wave frequency is reduction of size and mass of the system. As size of radio frequency (RF) components, including antennae gets smaller as its operation frequency increases, we can greatly minimize the bulk size of entire SAR instrument in millimeter wave region. The presented SAR instrument was integrated to (110 mm ×

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110 mm × 60 mm) sized casing. Mass of the instrument was measured to be less than 700 g. Important system parameters are summarized in Table 1.

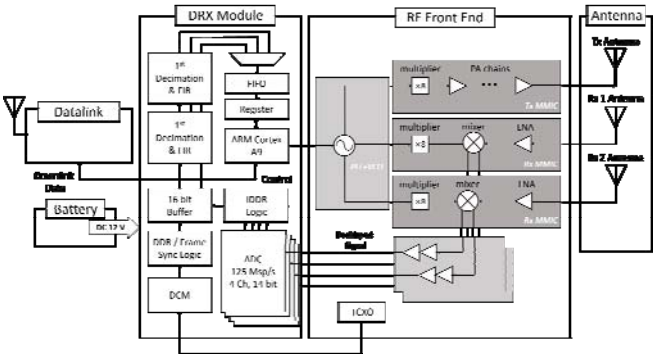


Fig. 2. Schematic Diagram of the System.

TABLE I. SYSTEM PARAMETERS

System Parameters	Values
Center Frequency	77 GHz
Bandwidth	1 GHz
Transmitted Power	20 dBm
Reciver Gain	35 dB
Antenna Gain	16.46 dBi
Beamwidth	9.6° [a], 24° [b]
Sampling Frequency	125 MHz
Power Consumption	20.8 Watts
Size	(110 mm ×110 mm × 60 mm)
Mass	700 grams

[a.] Elevation beamwidth

[b.] Azimuth beamwidth

Fig. 2 shows the schematics of the SAR instrument. RF front end has a single-input-multiple-output (SIMO) configuration with dual receiver channels aiming for applications where instant fixed-baseline interferometry is required. Baseband LFM waveform with 9.625 GHz center frequency and 125 MHz of sweep bandwidth is generated by the phase–locked loop + voltage controlled oscillator (PLL+VCO) chip inside the transceiver, which is fed to each transmitter and receiver channels. Transmitter upconverts baseband signal to W-band RF signal using a times-8 frequency multiplier. Solid state power amplifier (SSPA) chain with net gain of 25 dBm, amplifies the transmit signals to provide sufficient transmission power. Standard WR-10 waveguide is used to feed signal to antenna.

At receivers, signal received from antenna is amplified with low noise amplifier (LNA) prior to the dechirping process. Receivers contain frequency multipliers to generate a replica of transmitted signal. Dechirped signal generated by mixing reference signal with received signal [8], is I/Q modulated and amplified once more with a two-stage amplifier before being transferred to the digital end. Consequently, total receiver gain is maximum 38 dB. Both the transmitter and the receiver are modularized and packaged into MMICs. Modularization of each channel helps to improve isolation between channels and provides ease in multiplexing of channels. Gross power consumption of RF Front end is 16.8 W, powered by DC 12 V Li-Po battery as a power source.

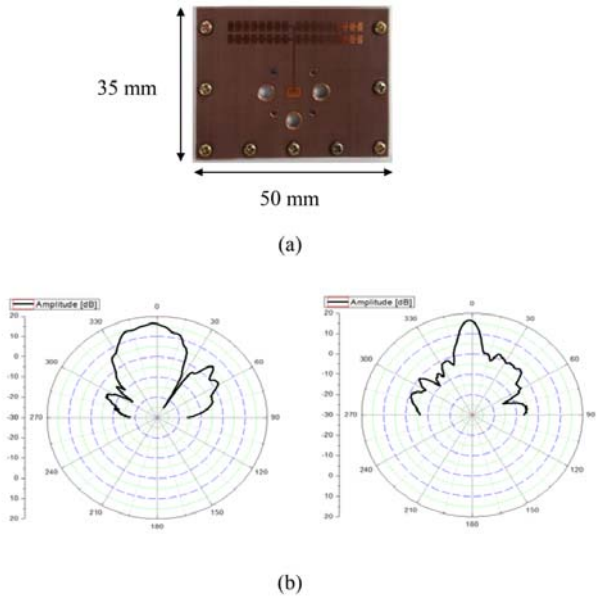


Fig. 3. (a) A fabricated copy of microstrip patch antenna integrated with waveguide transition. (b) Measured radiation pattern of microstrip patch antenna. From the left, azimuth and elevation – wise pattern respectively.

The presented system utilizes two types of antenna: A standard horn antenna and a microstrip patch antenna. Horn antenna is occasionally used for the evaluation purpose, however, we aim to use patch antenna for standard observation missions considering its' compact size and light weight. A decent microstrip patch antenna which suits with the system was designed and fabricated in cooperation with Korea Electronic Technology Institute (KETI). The array is consisted of (2 × 2) sub-arrays, where each sub-array is a linear array of 8 patches. The gain of antenna was measured to be maximum 16.46 dB at the radar's center frequency, with half power beamwidth of 9.6° along elevations. Sidelobe level was measured to be below -16 dB along elevation. A waveguide to patch transition that matches transceiver's waveguide port and antenna is designed as a whole. Transition is implemented with slot integrated waveguide to minimize the radiation loss and maintain the compactness of the system. With this transition, more than 2 GHz of bandwidth of where return loss is suppressed below -20 dB is secured.

B. Digital Receiver Module and Datalink

The digital receiver DRX module is implemented using a commercial FPGA board attached with an ARM Cortex CPU. Since power consumption increases as FPGA's degree of integration gets higher, Zynq XC7Z030 low power FPGA was chosen. In the system, DRX only performs two vital tasks only, data acquisition and frontend control, to keep DRX functionality to be simple as possible for the power criterion.

Four lanes of radar echo, received from I and Q channel of two receivers are fed to four channel parallel analog to

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digital converters (ADC) with 125 MHz of sampling frequency and 14 quantization bits, located at the frontmost path of DRX. Each quantized I/Q echo data stream is then transferred to the FPGA via low voltage differential signaling (LVDS). A chain of low pass filters and decimation filters implemented within the FPGA downsamples data stream to enhance the signal quality and to unburden datalink. To provide extra efficiency for downlink throughput, data streams are compressed after downsampling. Finally, streams of radar echo are parceled out pulse-wise and are sent to the datalink via gigabit ethernet. Along with the data acquisition, DRX deals with the control of sub-systems' operation. With CPU serving as a master for each sub-systems, it sends a proper control words for PLL+VCO and ADCs.

One of the sub-objective our system is to implement a real-time SAR image acquisition procedure via seamless data downlink. Therefore, we introduced the datalink to establish a high speed communication between the on-board system and the ground control. To maximize throughput, protocol between the on-board radar system and the ground control is implemented by UDP. This enables us to achieve higher data rate than standard TCP, since handshaking procedure is ignored in UDP. As a trade-off, there may be some missing pulses within the collected set of data which must be compensated for. When frequency of data loss is sparse enough, we compensate the loss with pulse interpolation or filling corresponding azimuth bin with zeros. When a large window of data is lost, energy loss for SAR data will be tremendous. This leads to the degradation of SAR image's quality. Hence, DRX board steadily backs up radar echoes to SDRAM buffer and internal storage integrated within itself, in preparation for the case where connection is completely lost for a long period of time. Data bandwidth of datalink is maximum 80 MHz. Connection range of the datalink stretches more than 1 km. Gross power consumption of DRX and datalink module is 8 W, powered by same power source as well as RF frontend.

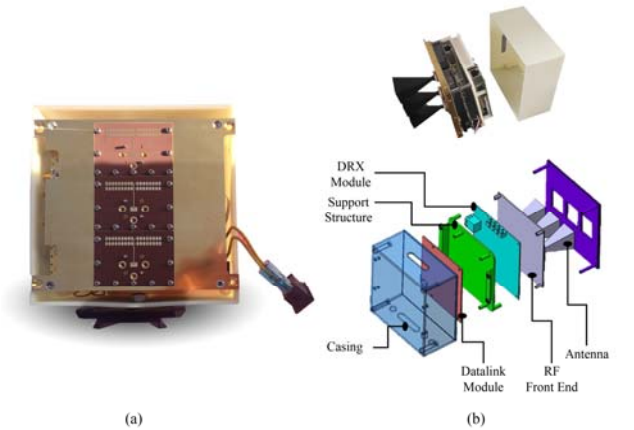


Fig. 4. (a) SAR instrument packaged and integrated with antennae (b) Structual layers of SAR instrument.

C. Multirotor UAV Platform

To minimize the non-ideal motion of the platform, the SAR instrument was mounted on an UAV equipped with a gimbal. A 6 rotor DJI Matrice 600™ UAV, with actively controlled Ronin™ gimbal manufactured by the same company, originally designed for the professional aerial photography is used for the SAR platform. Maximum take-off mass of platform that ensures stable flight is 15.5 kg. Since vehicle itself weighs 9.5 kg, weight budget for the payload is 6 kg. The total weight of SAR instrument, battery, datalink antenna, and gimbal is 4 kg, which is readily affordable to the platform.

The platform vehicle can fly in autopilot mode, which uses geographic coordinate positions provided by the user to move from one point to another. The autopilot mode uses a combined navigation data consisted of observations from GNSS real time kinematics (RTK) and accelerations acquired by differential inertial measurement unit. Adopting autopilot along with active gimbal can minimize the magnitude of motion and attitude errors. This benefits the system greatly, since the major drawback of using small rotor wing aircraft is the inherent instability of platform motion. Non-uniform platform velocity, range drift and Doppler centroid introduced by undesired squint are typical phenomena that occur as platform motion becomes non-ideal which introduce phase errors to the SAR data. The phase errors increase with the operating frequency making imaging in millimeter wave region challenging. Though it is possible to compensate motion errors by a certain degree with autofocus, it is more desirable to suppress them in the first place if possible. The platform of the presented system using RTK based navigation provides maximum 2 cm position accuracy, resulting in the decent control of flight path and velocity. Meanwhile, the gimbal suppresses deviations of both depression and squint angle into near zero level. Therefore, it is possible to acquire SAR data with small motion error using a small UAV platform.

III. Experiment

To validate that the system has a capability of acquiring proper data to synthesize a SAR image, we've conducted a simple airborne SAR imaging experiment. Details on the experiment, including its set-up, treatments we've done, and the result is presented in this section.

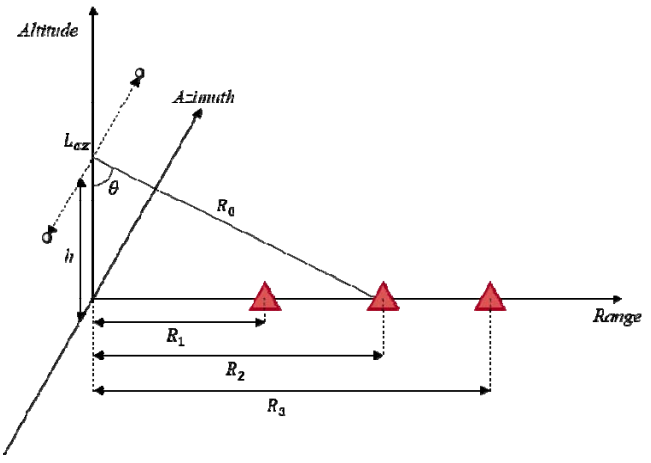


Fig. 5. The SAR Geometry of the experiment

A. Experimental Setup

The experiment is conducted as a broadside observation to maintain both analysis and signal processing simple and straightforward. Three trihedral corner reflectors with RCS of 100 m^2 at radar center frequency were used as targets.

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Each corner reflector is stationed at 52 m, 95 m, 136 m from the ground center of radar's path. SAR geometry parameters are roughly chosen, since the purpose of the experiment is to prove elementary SAR capability in arbitrary situation. The altitude of platform is set to 30 m, and slant range to the scene center is set to 100 m. This determines the depression angle to be 17°. The synthetic aperture length was chosen to be 8 m. To satisfy Nyquist criterion for spatial frequency, pulse repetition frequency (PRF) was selected to be 500 Hz. The velocity of platform vehicle was 2 m/s during entire flight. The geometry of the experiment is visualized in Fig. 5.

B. *Image Synthesis*

In FMCW radars, applying FFT over dechirped raw data will yield a frequency domain dual of range profile without additional treatments. Therefore, after FFT, proper focusing over azimuth-wise direction will produce a SAR image. Several pre-processing including range deskew, precise platform velocity estimation and autofocus can be applied to enhance the quality of image. In the experiment, deskewing and velocity estimation were skipped, since squint angle and azimuth-wise motion could be assumed to be well controlled. However, though motion errors were suppressed to few centimeters' degree, it is still several times greater than the radar's wavelength. Therefore, we applied phase gradient autofocus algorithm (PGA) to compensate phase errors within received signals [9]-[10]. Range migration algorithm (RMA) was chosen to focus a SAR image [11]. Other representative SAR algorithms like range-Doppler algorithm (RDA) or chirp scaling algorithm (CSA) have less or no benefit in this case.

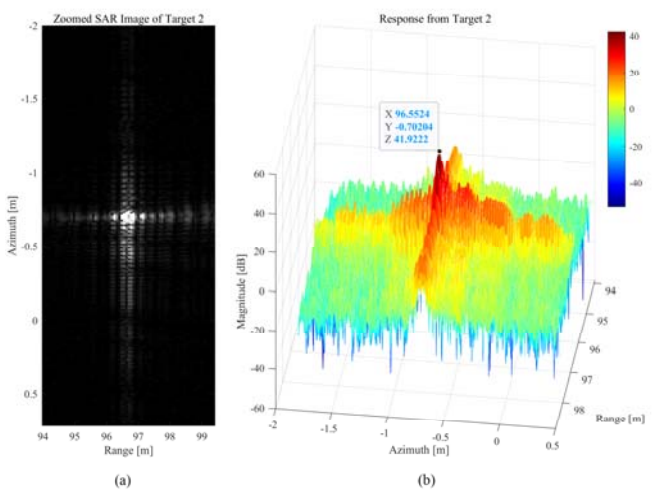


Fig. 6. (a) A slice of SAR image corresponding to the response from the target located at scene center (b) Mesh representation of same image slice for target response analysis.

C. *Result*

Echoes from the scene are embedded below echoes from the target since target RCS is far greater than ground reflectivity. Moreover, corner reflectors can be approximated to a point targets when their dimension is small enough. Therefore, we can regard the imaging of the target as an impulse response function (IRF) of the system. Fig. 6 shows the response from the second target, located at the ground range of 95 m, which corresponds to the scene center. Focusing result for the second target shows a profile that is fairly similar to a 2D–sinc function, which is a theoretical IRF for any SAR systems. Peak magnitude of the response is observed to be 42 dB. Sidelobe levels for both range and azimuth axis are roughly around 24 dB in average. The rule of thumb suggests that more than 13 dB difference between the peak and sidelobe is required to synthesize a decent SAR image. Hence, it is plausible to state that the system is capable of acquiring proper SAR data. Responses from other reflectors are deliberately omitted, since those targets were residing at the perimeter of the expected SAR image swath, therefore the degree of focusing was rather non-optimal, compared to that from the scene center.

IV. Conclusion

In this paper, design and implementation of SAR system which aims to be applied for drone based operations is presented. The system adopts a 77 GHz W–band architecture to achieve a decent compactness and efficiency required to compensate restrictions arising from the small carrying capacity of drones. SIMO configuration of dual receivers allows the system a wider range of operation including a single path interferometry. Precise navigation and attitude control provided by the multi rotor platform vehicle, the system can collect a decent SAR data. To prove this concept, a simple experiment of airborne SAR imaging was conducted. With corner reflector as targets, IRF of the proposed system was acquired. Analysis on the magnitude of IRF provides an evidence that a decent SAR data can be collected upon small drones, even in the millimeter wave region.

For future work, we will further extend the system's capability by optimizing hardware and enhancing overall system integrity. Extensive experiments to measure the performance of the presented system in more advanced operational modes, including image acquisition in wide stripmap mode and single-path cross-/along-track InSAR mode, are planned to be conducted once hardware revision is completed. Simultaneously, a research to validate the effectiveness of using drone based SAR systems in detecting structural hazard within the local scene will be proceeded as well.

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